

COER UNIVERSITY

ROORKEE

Department of Computer Science & Engineering

LABORATORY FILE

Subject	Full Stack Development
Experiment No.	01
Experiment Title	Building the Storefront (HTML + CSS)
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Experiment No. 01

Title: Building the Storefront (HTML + CSS)

Aim

To design and develop the static page structure of a mini e-commerce storefront using HTML and style it into a responsive layout using CSS, covering the home page, product listing, product detail, cart and checkout pages.

Theory

HTML (HyperText Markup Language) provides the structural foundation of web pages through elements such as headers, navigation, sections, tables and forms. CSS (Cascading Style Sheets) is used to control presentation, layout and responsiveness. In this experiment, Flexbox and CSS Grid are applied for navigation and product card arrangement. CSS custom properties (variables) define a consistent colour theme. Media queries ensure the layout adapts correctly on mobile-width screens. Together, HTML and CSS create a complete static storefront without any backend or JavaScript.

Software Required

- Visual Studio Code (or any text editor)
- Web Browser (Google Chrome / Microsoft Edge)

Procedure

1. Create a project folder named gadgethub-frontend.
2. Create five HTML files: index.html, products.html, product-detail.html, cart.html and checkout.html.
3. Add a common header with store logo and navigation links (Home, Products, Cart) on every page.
4. Build the content of each page as specified (hero section, product cards, detail view, cart table, checkout form).
5. Create style.css with CSS variables, Flexbox navbar, product card styles, CSS Grid layout, form styling and a media query for mobile.
6. Link style.css to all HTML pages and open the pages in a browser to verify layout and navigation.
7. Test responsiveness by resizing the browser window.

Source Code

1. index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>GadgetHub | Home</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <!-- Common Header -->
  <header>
    <nav class="navbar">
      <a href="index.html" class="logo">Gadget&ltspan>Hub</span></a>
      <ul class="nav-links">
        <li><a href="index.html" class="active">Home</a></li>
        <li><a href="products.html">Products</a></li>
        <li><a href="cart.html">Cart</a></li>
      </ul>
    </nav>
  </header>

  <main>
    <!-- Hero / Banner Section -->
    <section class="hero">
      <h1>Welcome to GadgetHub</h1>
      <p>Your one-stop destination for the latest tech gadgets, accessories, and smart devices at unbeatable prices.</p>
      <a href="products.html" class="btn">Shop Now</a>
    </section>

    <!-- Featured Products -->
    <section>
      <h2 class="section-title">Featured Products</h2>
      <div class="product-grid">
        <!-- Product 1 -->
        <div class="product-card">
          
          <div class="card-body">
            <h3>Wireless Earbuds Pro</h3>
            <p class="price">₹2,499</p>
            <a href="product-detail.html" class="btn">View Details</a>
          </div>
        </div>
        <!-- Product 2 -->
        <div class="product-card">
          
          <div class="card-body">
            <h3>Smart Watch X5</h3>
            <p class="price">₹4,999</p>
            <a href="product-detail.html" class="btn">View Details</a>
          </div>
        </div>
        <!-- Product 3 -->
```

```
<div class="product-card">
  
  <div class="card-body">
    <h3>Portable Bluetooth Speaker</h3>
    <p class="price">₹1,799</p>
    <a href="product-detail.html" class="btn">View Details</a>
  </div>
</div>
<!-- Product 4 -->
<div class="product-card">
  
  <div class="card-body">
    <h3>20000mAh Power Bank</h3>
    <p class="price">₹1,299</p>
    <a href="product-detail.html" class="btn">View Details</a>
  </div>
</div>
</div>
</section>
</main>

<!-- Common Footer -->
<footer>
  <p><strong>GadgetHub</strong> – Your Trusted Tech Store</p>
  <p>Email: support@gadgethub.com | Phone: +91 98765 43210</p>
  <p>&copy; 2026 GadgetHub. All rights reserved.</p>
</footer>
</body>
</html>
```

2. products.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>GadgetHub | Products</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <nav class="navbar">
      <a href="index.html" class="logo">Gadget<span>Hub</span></a>
      <ul class="nav-links">
        <li><a href="index.html">Home</a></li>
        <li><a href="products.html" class="active">Products</a></li>
        <li><a href="cart.html">Cart</a></li>
      </ul>
    </nav>
  </header>

  <main>
    <h2 class="section-title">All Products</h2>
```

```
<div class="product-grid">
  <!-- Product 1 -->
  <div class="product-card">
    
    <div class="card-body">
      <h3>Wireless Earbuds Pro</h3>
      <p class="price">₹2,499</p>
      <button class="btn">Add to Cart</button>
    </div>
  </div>
  <!-- Product 2 -->
  <div class="product-card">
    
    <div class="card-body">
      <h3>Smart Watch X5</h3>
      <p class="price">₹4,999</p>
      <button class="btn">Add to Cart</button>
    </div>
  </div>
  <!-- Product 3 -->
  <div class="product-card">
    
    <div class="card-body">
      <h3>Portable Bluetooth Speaker</h3>
      <p class="price">₹1,799</p>
      <button class="btn">Add to Cart</button>
    </div>
  </div>
  <!-- Product 4 -->
  <div class="product-card">
    
    <div class="card-body">
      <h3>20000mAh Power Bank</h3>
      <p class="price">₹1,299</p>
      <button class="btn">Add to Cart</button>
    </div>
  </div>
  <!-- Product 5 -->
  <div class="product-card">
    
    <div class="card-body">
      <h3>7-in-1 USB-C Hub</h3>
      <p class="price">₹2,199</p>
      <button class="btn">Add to Cart</button>
    </div>
  </div>
  <!-- Product 6 -->
  <div class="product-card">
    
    <div class="card-body">
      <h3>Ergonomic Wireless Mouse</h3>
      <p class="price">₹899</p>
      <button class="btn">Add to Cart</button>
    </div>
  </div>
  <!-- Product 7 -->
  <div class="product-card">
    
```

```
<div class="card-body">
  <h3>Aluminum Laptop Stand</h3>
  <p class="price">₹1,499</p>
  <button class="btn">Add to Cart</button>
</div>
</div>
<!-- Product 8 -->
<div class="product-card">
  
  <div class="card-body">
    <h3>RGB Mechanical Keyboard</h3>
    <p class="price">₹3,499</p>
    <button class="btn">Add to Cart</button>
  </div>
</div>
</div>
</main>

<footer>
  <p><strong>GadgetHub</strong> – Your Trusted Tech Store</p>
  <p>Email: support@gadgethub.com | Phone: +91 98765 43210</p>
  <p>&copy; 2026 GadgetHub. All rights reserved.</p>
</footer>
</body>
</html>
```

3. product-detail.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>GadgetHub | Product Detail</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <nav class="navbar">
      <a href="index.html" class="logo">Gadget<span>Hub</span></a>
      <ul class="nav-links">
        <li><a href="index.html">Home</a></li>
        <li><a href="products.html">Products</a></li>
        <li><a href="cart.html">Cart</a></li>
      </ul>
    </nav>
  </header>

  <main>
    <div class="product-detail">
      <div>
        
      </div>
      <div class="detail-info">
        <h1>Wireless Earbuds Pro</h1>
        <p class="price">₹2,499</p>
```

```
<p>
  Experience premium sound with our Wireless Earbuds Pro. Featuring active noise cancellation,
  30-hour battery life with the charging case, IPX5 water resistance, and crystal-clear calls.
  Perfect for music, workouts, and everyday use. Compatible with both Android and iOS devices.
</p>
<div class="quantity-selector">
  <label for="qty">Quantity:</label>
  <input type="number" id="qty" name="qty" value="1" min="1" max="10">
</div>
<button class="btn">Add to Cart</button>
</div>
</div>
</main>

<footer>
  <p><strong>GadgetHub</strong> – Your Trusted Tech Store</p>
  <p>Email: support@gadgethub.com | Phone: +91 98765 43210</p>
  <p>&copy; 2026 GadgetHub. All rights reserved.</p>
</footer>
</body>
</html>
```

4. cart.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>GadgetHub | Cart</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <nav class="navbar">
      <a href="index.html" class="logo">Gadget<span>Hub</span></a>
      <ul class="nav-links">
        <li><a href="index.html">Home</a></li>
        <li><a href="products.html">Products</a></li>
        <li><a href="cart.html" class="active">Cart</a></li>
      </ul>
    </nav>
  </header>

  <main>
    <h2 class="section-title">Your Shopping Cart</h2>
    <table class="cart-table">
      <thead>
        <tr>
          <th>Item Name</th>
          <th>Quantity</th>
          <th>Price</th>
          <th>Subtotal</th>
        </tr>
```

```
</thead>
<tbody>
  <tr>
    <td>Wireless Earbuds Pro</td>
    <td>1</td>
    <td>₹2,499</td>
    <td>₹2,499</td>
  </tr>
  <tr>
    <td>Smart Watch X5</td>
    <td>1</td>
    <td>₹4,999</td>
    <td>₹4,999</td>
  </tr>
  <tr>
    <td>Portable Bluetooth Speaker</td>
    <td>2</td>
    <td>₹1,799</td>
    <td>₹3,598</td>
  </tr>
  <tr class="total-row">
    <td colspan="3">Grand Total</td>
    <td>₹11,096</td>
  </tr>
</tbody>
</table>

<div class="cart-actions">
  <a href="products.html" class="btn" style="background:#5f6368;">Continue Shopping</a>
  <a href="checkout.html" class="btn">Proceed to Checkout</a>
</div>
</main>

<footer>
  <p><strong>GadgetHub</strong> – Your Trusted Tech Store</p>
  <p>Email: support@gadgethub.com | Phone: +91 98765 43210</p>
  <p>&copy; 2026 GadgetHub. All rights reserved.</p>
</footer>
</body>
</html>
```

5. checkout.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>GadgetHub | Checkout</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <nav class="navbar">
      <a href="index.html" class="logo">Gadget<span>Hub</span></a>
      <ul class="nav-links">
```

```
        <li><a href="index.html">Home</a></li>
        <li><a href="products.html">Products</a></li>
        <li><a href="cart.html">Cart</a></li>
    </ul>
</nav>
</header>

<main>
    <h2 class="section-title">Checkout</h2>
    <form class="checkout-form" action="#" method="post">
        <div class="form-group">
            <label for="name">Full Name</label>
            <input type="text" id="name" name="name" placeholder="Enter your full name" required>
        </div>

        <div class="form-group">
            <label for="address">Address</label>
            <input type="text" id="address" name="address" placeholder="House No., Street, Area" required>
        </div>

        <div class="form-group">
            <label for="city">City</label>
            <input type="text" id="city" name="city" placeholder="Enter city" required>
        </div>

        <div class="form-group">
            <label for="pincode">Pincode</label>
            <input type="text" id="pincode" name="pincode" placeholder="6-digit pincode" pattern="[0-9]{6}" required>
        </div>

        <div class="form-group">
            <label for="phone">Phone Number</label>
            <input type="tel" id="phone" name="phone" placeholder="10-digit mobile number" pattern="[0-9]{10}"
required>
        </div>

        <div class="form-group">
            <label>Payment Method</label>
            <div class="radio-group">
                <label>
                    <input type="radio" name="payment" value="cod" checked> Cash on Delivery
                </label>
                <label>
                    <input type="radio" name="payment" value="upi"> UPI
                </label>
                <label>
                    <input type="radio" name="payment" value="card"> Credit / Debit Card
                </label>
            </div>
        </div>

        <button type="submit" class="btn">Place Order</button>
    </form>
</main>
```

```
<footer>
  <p><strong>GadgetHub</strong> – Your Trusted Tech Store</p>
  <p>Email: support@gadgethub.com | Phone: +91 98765 43210</p>
  <p>&copy; 2026 GadgetHub. All rights reserved.</p>
</footer>
</body>
</html>
```

6. style.css

```
/* =====
GadgetHub - Mini E-Commerce Storefront
style.css | Full Stack Development Lab
===== */

/* ----- CSS Custom Properties (Theme) ----- */
:root {
  --primary-color: #1a73e8;      /* Main brand blue */
  --accent-color: #ff6d00;       /* Accent orange */
  --dark-color: #202124;        /* Near-black text */
  --light-bg: #f8f9fa;          /* Light background */
  --card-shadow: 0 4px 12px rgba(0, 0, 0, 0.08);
  --radius: 10px;
  --transition: all 0.25s ease;
}

/* ----- Reset & Base ----- */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  color: var(--dark-color);
  background-color: var(--light-bg);
  line-height: 1.6;
  min-height: 100vh;
  display: flex;
  flex-direction: column;
}

a {
  text-decoration: none;
  color: inherit;
}

img {
  max-width: 100%;
  display: block;
}

/* ----- Header & Navbar (Flexbox) ----- */
```

```
header {  
  background-color: #ffffff;  
  box-shadow: 0 2px 8px rgba(0, 0, 0, 0.06);  
  position: sticky;  
  top: 0;  
  z-index: 100;  
}
```

```
.navbar {  
  max-width: 1200px;  
  margin: 0 auto;  
  padding: 0.9rem 1.5rem;  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
  flex-wrap: wrap;  
  gap: 1rem;  
}
```

```
.logo {  
  font-size: 1.6rem;  
  font-weight: 700;  
  color: var(--primary-color);  
  letter-spacing: -0.5px;  
}
```

```
.logo span {  
  color: var(--accent-color);  
}
```

```
.nav-links {  
  display: flex;  
  gap: 1.8rem;  
  list-style: none;  
}
```

```
.nav-links a {  
  font-weight: 500;  
  color: var(--dark-color);  
  transition: var(--transition);  
  padding: 0.3rem 0;  
  position: relative;  
}
```

```
.nav-links a:hover,  
.nav-links a.active {  
  color: var(--primary-color);  
}
```

```
.nav-links a.active::after {  
  content: '';  
  position: absolute;  
  bottom: -4px;  
  left: 0;  
  width: 100%;  
}
```

```
height: 2px;
background: var(--primary-color);
}

/* ----- Main Content ----- */
main {
  flex: 1;
  max-width: 1200px;
  margin: 0 auto;
  padding: 2rem 1.5rem;
  width: 100%;
}

/* ----- Hero Section ----- */
.hero {
  background: linear-gradient(135deg, var(--primary-color), #0d47a1);
  color: white;
  border-radius: var(--radius);
  padding: 3.5rem 2rem;
  text-align: center;
  margin-bottom: 3rem;
}

.hero h1 {
  font-size: 2.4rem;
  margin-bottom: 0.8rem;
}

.hero p {
  font-size: 1.15rem;
  opacity: 0.9;
  margin-bottom: 1.5rem;
  max-width: 600px;
  margin-left: auto;
  margin-right: auto;
}

.btn {
  display: inline-block;
  padding: 0.75rem 1.8rem;
  background-color: var(--accent-color);
  color: white;
  border: none;
  border-radius: 6px;
  font-weight: 600;
  cursor: pointer;
  transition: var(--transition);
  font-size: 1rem;
}

.btn:hover {
  background-color: #e65100;
  transform: translateY(-2px);
  box-shadow: 0 4px 12px rgba(255, 109, 0, 0.35);
}
```

```
.btn-outline {
  background: transparent;
  border: 2px solid white;
  color: white;
}

.btn-outline:hover {
  background: white;
  color: var(--primary-color);
}

/* ----- Section Titles ----- */
.section-title {
  font-size: 1.7rem;
  margin-bottom: 1.5rem;
  color: var(--dark-color);
  position: relative;
  padding-bottom: 0.5rem;
}

.section-title::after {
  content: '';
  position: absolute;
  bottom: 0;
  left: 0;
  width: 60px;
  height: 3px;
  background: var(--accent-color);
  border-radius: 2px;
}

/* ----- Product Cards (Box Model) ----- */
.product-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(240px, 1fr));
  gap: 1.5rem;
}

.product-card {
  background: white;
  border-radius: var(--radius);
  box-shadow: var(--card-shadow);
  overflow: hidden;
  transition: var(--transition);
  display: flex;
  flex-direction: column;
}

.product-card:hover {
  transform: translateY(-6px);
  box-shadow: 0 10px 24px rgba(0, 0, 0, 0.12);
}
```

```
.product-card img {
  width: 100%;
  height: 180px;
  object-fit: cover;
  background: #e8eaed;
}

.product-card .card-body {
  padding: 1.1rem;
  flex: 1;
  display: flex;
  flex-direction: column;
}

.product-card h3 {
  font-size: 1.05rem;
  margin-bottom: 0.4rem;
  color: var(--dark-color);
}

.product-card .price {
  font-size: 1.2rem;
  font-weight: 700;
  color: var(--primary-color);
  margin-bottom: 0.8rem;
}

.product-card .btn {
  margin-top: auto;
  width: 100%;
  text-align: center;
  padding: 0.6rem;
  font-size: 0.95rem;
}

/* ----- Product Detail Page ----- */
.product-detail {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 2.5rem;
  background: white;
  padding: 2rem;
  border-radius: var(--radius);
  box-shadow: var(--card-shadow);
}

.product-detail img {
  width: 100%;
  border-radius: var(--radius);
  background: #e8eaed;
  height: 380px;
  object-fit: cover;
}

.detail-info h1 {
```

```
    font-size: 1.9rem;
    margin-bottom: 0.6rem;
}

.detail-info .price {
    font-size: 1.6rem;
    font-weight: 700;
    color: var(--primary-color);
    margin-bottom: 1rem;
}

.detail-info p {
    color: #5f6368;
    margin-bottom: 1.5rem;
}

.quantity-selector {
    display: flex;
    align-items: center;
    gap: 0.8rem;
    margin-bottom: 1.5rem;
}

.quantity-selector label {
    font-weight: 500;
}

.quantity-selector input {
    width: 70px;
    padding: 0.5rem;
    border: 1px solid #dadce0;
    border-radius: 6px;
    text-align: center;
    font-size: 1rem;
}

/* ----- Cart Table ----- */
.cart-table {
    width: 100%;
    border-collapse: collapse;
    background: white;
    border-radius: var(--radius);
    overflow: hidden;
    box-shadow: var(--card-shadow);
}

.cart-table th,
.cart-table td {
    padding: 1rem 1.2rem;
    text-align: left;
    border-bottom: 1px solid #e8eaed;
}

.cart-table th {
```

```
    background: var(--primary-color);
    color: white;
    font-weight: 600;
}

.cart-table tr:last-child td {
    border-bottom: none;
}

.cart-table .total-row {
    font-weight: 700;
    background: #f1f3f4;
}

.cart-table .total-row td {
    font-size: 1.15rem;
    color: var(--primary-color);
}

.cart-actions {
    margin-top: 1.5rem;
    display: flex;
    justify-content: flex-end;
    gap: 1rem;
}

/* ----- Checkout Form ----- */
.checkout-form {
    background: white;
    padding: 2rem;
    border-radius: var(--radius);
    box-shadow: var(--card-shadow);
    max-width: 600px;
    margin: 0 auto;
}

.form-group {
    margin-bottom: 1.2rem;
}

.form-group label {
    display: block;
    font-weight: 500;
    margin-bottom: 0.4rem;
    color: var(--dark-color);
}

.form-group input,
.form-group select,
.form-group textarea {
    width: 100%;
    padding: 0.7rem 0.9rem;
    border: 1px solid #dadce0;
    border-radius: 6px;
```

```
font-size: 1rem;
transition: var(--transition);
}

.form-group input:focus,
.form-group select:focus,
.form-group textarea:focus {
  outline: none;
  border-color: var(--primary-color);
  box-shadow: 0 0 3px rgba(26, 115, 232, 0.15);
}

.radio-group {
  display: flex;
  gap: 1.5rem;
  flex-wrap: wrap;
  margin-top: 0.4rem;
}

.radio-group label {
  display: flex;
  align-items: center;
  gap: 0.4rem;
  font-weight: 400;
  cursor: pointer;
}

.checkout-form .btn {
  width: 100%;
  margin-top: 0.8rem;
  padding: 0.9rem;
  font-size: 1.05rem;
}

/* ----- Footer ----- */
footer {
  background: var(--dark-color);
  color: #e8eaed;
  text-align: center;
  padding: 1.8rem 1.5rem;
  margin-top: 3rem;
}

footer p {
  margin: 0.3rem 0;
  font-size: 0.95rem;
}

footer a {
  color: var(--accent-color);
}

/* ----- Responsive Design (@media) ----- */
@media (max-width: 768px) {
```

```
.navbar {
  flex-direction: column;
  align-items: flex-start;
}

.nav-links {
  flex-direction: column;
  gap: 0.8rem;
  width: 100%;
}

.hero h1 {
  font-size: 1.8rem;
}

.product-grid {
  grid-template-columns: 1fr;
}

.product-detail {
  grid-template-columns: 1fr;
}

.product-detail img {
  height: 260px;
}

.cart-table th,
.cart-table td {
  padding: 0.7rem 0.6rem;
  font-size: 0.9rem;
}

.cart-actions {
  flex-direction: column;
}

.cart-actions .btn {
  width: 100%;
}
}

@media (max-width: 480px) {
  .hero {
    padding: 2.2rem 1.2rem;
  }

  main {
    padding: 1.2rem 1rem;
  }
}
```

Output

The following screenshots show the final webpage output after linking the CSS file and opening the pages in a browser.

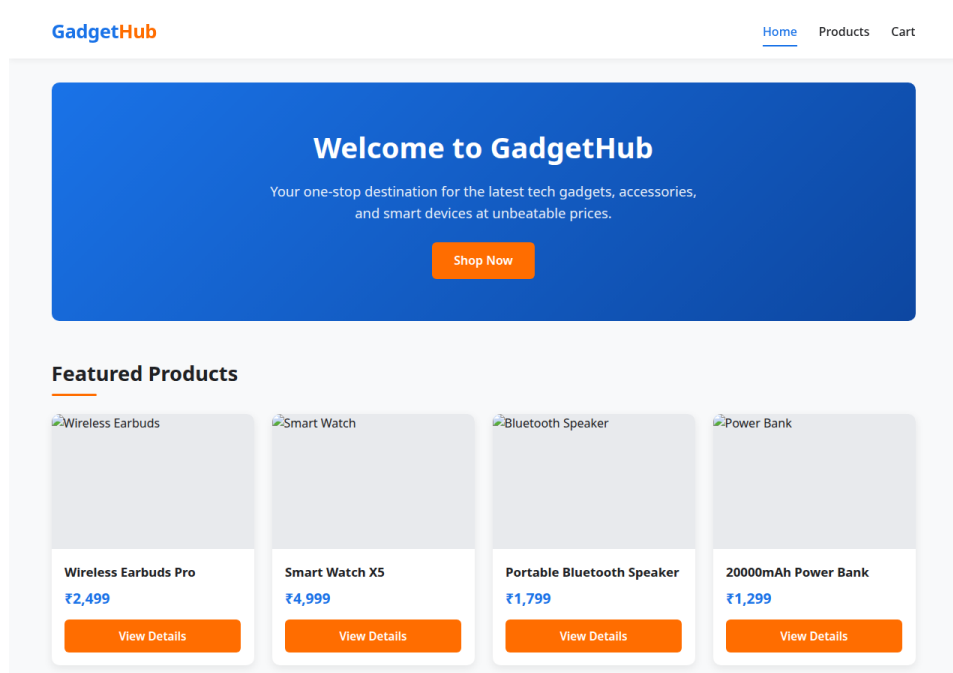


Figure 1: Final Webpage Output – Home Page (index.html)

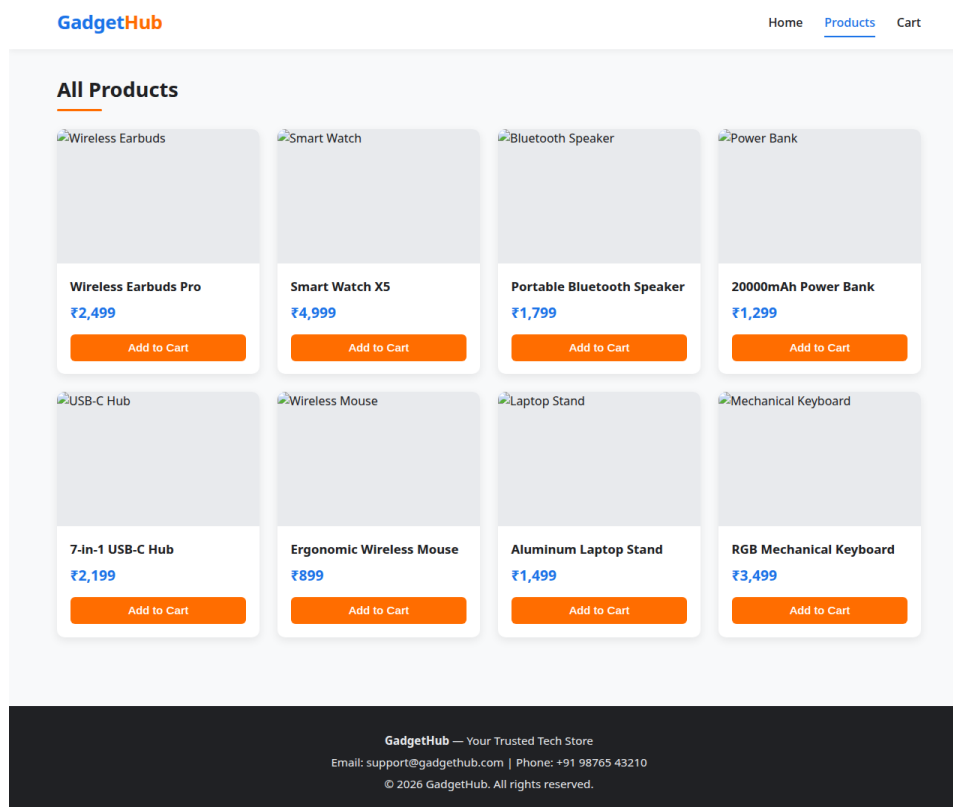


Figure 2: Final Webpage Output – Products Listing Page

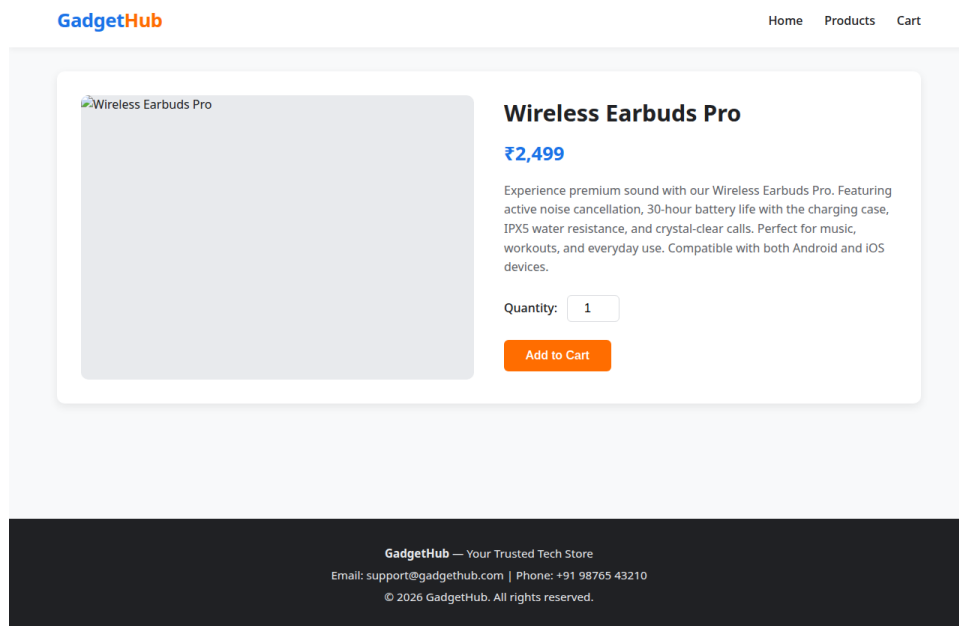


Figure 3: Final Webpage Output – Product Detail Page

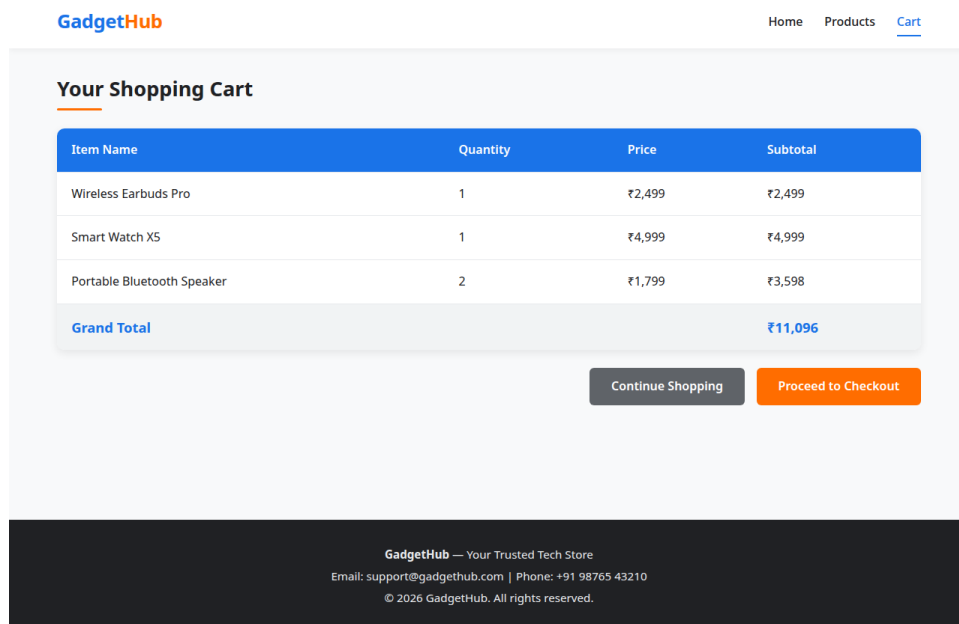


Figure 4: Final Webpage Output – Shopping Cart Page

GadgetHub Home Products Cart

Checkout

Full Name
Enter your full name

Address
House No., Street, Area

City
Enter city

Pincode
6-digit pincode

Phone Number
10-digit mobile number

Payment Method

☒ Cash on Delivery ☐ UPI ☐ Credit / Debit Card

Place Order

GadgetHub — Your Trusted Tech Store
Email: support@gadgethub.com | Phone: +91 98765 43210
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Figure 5: Final Webpage Output – Checkout Form Page

Result

A complete static and responsive e-commerce storefront was successfully developed using HTML and CSS. All five pages (Home, Products, Product Detail, Cart and Checkout) are linked through a common navigation bar, styled with a consistent theme, and adapt correctly to mobile screen widths.